EMT Disconnected One-Point Building Blocks

Quark Loops, Gluon Loops, Stochastic Sources, and HDF5 Outputs

PyQUDA Measurement Notes

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1 Purpose and File Map

This note documents the hadron-independent one-point functions used to build disconnected contributions to proton and pion energy-momentum tensor (EMT) matrix elements. The main implementation is

`pyquda_measurement_utils/Disconnected_1pt_EMT_vibe_develop.py`.

The Perlmutter application entry points are in

`application/EMT_disconnected_1pt/perlmutter`.

The shorter command-oriented workflow guide is

`docs/EMT_disconnected_1pt/EMT_proton_disconnected_guide.md`.

The one-point workflows do not form hadron three-point functions by themselves. They measure vacuum loop building blocks. A disconnected hadron matrix element is formed later by correlating these one-point loops with hadron two-point functions on the same gauge ensemble.

This document describes the current `Pyquda_Measurement_develop` checkout. Operational examples and probing plans are in `application/EMT_disconnected_1pt/PROBING.md`. Old data must be read with their matching original checkout; they are not migrated by develop. Historical numerical checks below are identified explicitly and are not GPU/MPI validation of the new probing implementation.

2 Conventions

We work in Euclidean spacetime with four directions. Mathematical formulas use $\mu, \nu, \rho = 1, 2, 3, 4$, while Python loops use zero-based indices $\mu, \nu, \rho = 0, 1, 2, 3$. The temporal direction is the fourth Euclidean direction. PyQUDA stores local lattice fields in even-odd layout; the displayed formulas suppress this storage detail.

Spatial phases use `MomentumPhase.getPhases`. For spatial momentum transfer q, this note writes the phase as

$$\Phi_q(x) = e^{i\mathbf{q}\cdot\mathbf{x}}, \tag{1}$$

where only the spatial coordinates are Fourier transformed and the time coordinate t is kept explicit.

The spatial volume normalization used in the averaged HDF5 datasets is

$$V_3 = L_x L_y L_z. \tag{2}$$

Optional raw per-source quark datasets and production shard means are stored before this V_3 normalization. Finalized quark complete-base values, their averages and SEM, and averaged gluon datasets are volume-normalized. Do not divide finalized complete-base values by V_3 again.

3 Quark One-Point Workflow

The quark workflow is usually the first part to study. It contains the stochastic trace estimator, the hierarchical-probing source bookkeeping, the χ-type diagnostics, and the flowed quark EMT loop used in disconnected analysis.

3.1 Target Trace

A disconnected quark loop has the schematic form

$$L_{\Gamma}^{q}(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\ \mathrm{Tr}_{sc}\big[\Gamma[U(t_f)](x)\, D^{-1}[U](x,x)\big]\,, \tag{3}$$

where D^{-1} is the quark propagator, Γ is the local operator kernel, and Tr_{sc} denotes spin-color trace. Directly computing all diagonal entries of $D^{-1}(x,x)$ is too expensive, so the trace is estimated with stochastic sources.

3.2 Stochastic Trace Estimator

Let ξ_r be a stochastic source vector whose spin-color-site components satisfy

$$\langle \xi_r \rangle_{\mathrm{noise}} = 0, \qquad \Big\langle \xi_r(i)\, \xi_r^{\dagger}(j) \Big\rangle_{\mathrm{noise}} = \delta_{ij}. \tag{4}$$

Here i, j are combined site, spin, and color indices. The code obtains

$$\eta_r = D^{-1}\xi_r \tag{5}$$

from the Dirac inverter. For any operator matrix A,

$$\Big\langle \xi_r^{\dagger} A \eta_r \Big\rangle_{\mathrm{noise}} = \Big\langle \xi_r^{\dagger} A D^{-1} \xi_r \Big\rangle_{\mathrm{noise}} \tag{6}$$

$$= \sum_{i,j,k} A_{ij}\, (D^{-1})_{jk}\, \Big\langle \xi_{r,i}^{\dagger} \xi_{r,k} \Big\rangle_{\mathrm{noise}} \tag{7}$$

$$= \sum_{i,j} A_{ij}\, (D^{-1})_{ji} = \mathrm{Tr}\big(AD^{-1}\big). \tag{8}$$

This is the Hutchinson trace-estimator logic used by the quark one-point workflow.

3.3 Z_n Noise

The ordinary stochastic scheme is selected with

$$\texttt{noise_scheme = zn}. \tag{9}$$

Each component of ξ is drawn independently from

$$\xi(i) \in \left\{ \exp\!\left(\frac{2\pi i r}{n}\right) \;\middle|\; r = 0, 1, \ldots, n-1 \right\}, \tag{10}$$

where $n = \texttt{n_zn}$. The EMT quark one-point workflow now defaults to full-volume $\mathbb{Z}_4$ noise. Its counter-based generator hashes the global four-dimensional coordinate, spin, color, configuration number, base-noise index, and optional stream salt. Consequently the global source is reproducible and independent of the chosen MPI decomposition.

This counter construction fixes a serious failure mode of rank-local backend random-number generators. If every MPI rank resets an ordinary generator to the same state s, ranks with the same local lattice shape draw the same array. For a rank-local index ℓ,

$$\xi_r(\ell) = F(s,\ell) = \xi_{r'}(\ell), \qquad r \neq r'. \tag{11}$$

The corresponding distinct global sites $x_r + \ell$ and $x_{r'} + \ell$ then have unit rather than vanishing noise covariance,

$$\mathbb{E}_{\xi}[\xi^*(x_r + \ell)\xi(x_{r'} + \ell)] = 1, \qquad r \neq r', \tag{12}$$

so the required identity in Eq. (4) is not sampled. This creates rank-periodic off-diagonal terms, can bias the stochastic trace, and changes the global source when the MPI decomposition changes. Mixing the rank into the seed removes exact repetition only for one fixed process layout; it still makes the source depend on the rank assignment. The production counter instead uses physical global coordinates and semantic keys, so the same configuration, base index, and stream reconstruct exactly the same global field for every MPI geometry. Legacy backend-RNG loop data are therefore not combined with counter-noise production data.

3.4 Quark EMT Operator Implemented in Code

The quark one-point kernel now measures the complete sixteen-element PyQUDA Dirac basis Γ_A. For each source it saves the local and one-derivative primitive bilinears

$$L_A^{(r)}(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\, \xi_r^\dagger(x;t_f)\Gamma_A \eta_r(x;t_f), \tag{13}$$

$$A_{A\mu}^{(r)}(q,t;t_f) = \frac{1}{2}\sum_{\mathbf{x}} \Phi_q(x)\, \xi_r^\dagger(x;t_f)\Gamma_A [D_{+\mu} - D_{-\mu}]\eta_r(x;t_f), \tag{14}$$

$$L_{A\mu}^{D,(r)}(q,t;t_f) = -\frac{1}{2}\left[A_{A\mu}^{(r)}(q,t;t_f) - A_{A^\sharp\mu}^{(r)}(-q,t;t_f)^*\right]. \tag{15}$$

Here $\Gamma_A^\sharp = \gamma_5\Gamma_A^\dagger\gamma_5$. Thus the stored derivative primitive is the complete two-sided operator including one closed fermion-loop Wick minus; A is only an internal right-acting intermediate and is not written to HDF5. The production Fourier phase is $\Phi_q(x) = \exp[+2\pi i\, \mathbf{q}\cdot(\mathbf{x}-\mathbf{o})/\mathbf{L}]$. The source-origin conversion used in three-point analysis consequently carries the opposite, negative phase. The gamma axis is ordered as

`5,T,T5,X,X5,Y,Y5,Z,Z5,I,SXT,SXY,SXZ,SYT,SYZ,SZT`

and the derivative axis is `X,Y,Z,T`. The sixteen gamma contractions are batched after each derivative is constructed, so this extension changes contraction and I/O cost but does not add inversions, fermion-flow updates, or covariant-derivative applications.

All EMT programs import this basis from

`pyquda_measurement_utils/fermion_bilinear_basis.py`.

In the DeGrand–Rossi representation returned by the current PyQUDA implementation, the four Euclidean generators are

$$\gamma_1 = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix}, \qquad \gamma_2 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \tag{16}$$

$$\gamma_3 = \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & -i \\ -i & 0 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}, \qquad \gamma_4 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \tag{17}$$

and

$$\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4 = \mathrm{diag}(1,1,-1,-1). \tag{18}$$

The resulting HDF5 Gamma axis is not numerical-ID order:

Index	Label	Raw matrix	Index	Label	Raw matrix
0	`5`	γ_5	8	`Z5`	$\gamma_3\gamma_5$
1	`T`	γ_4	9	`I`	$\mathbf{1}$
2	`T5`	$-\gamma_4\gamma_5$	10	`SXT`	$\gamma_1\gamma_4$
3	`X`	γ_1	11	`SXY`	$\gamma_1\gamma_2$
4	`X5`	$\gamma_1\gamma_5$	12	`SXZ`	$\gamma_1\gamma_3$
5	`Y`	γ_2	13	`SYT`	$\gamma_2\gamma_4$
6	`Y5`	$-\gamma_2\gamma_5$	14	`SYZ`	$\gamma_2\gamma_3$
7	`Z`	γ_3	15	`SZT`	$\gamma_3\gamma_4$

The corresponding PyQUDA IDs are

$$[15, 8, 7, 1, 14, 2, 13, 4, 11, 0, 9, 3, 5, 10, 6, 12]. \tag{19}$$

The numerical matrices themselves are stored in `gamma_matrices`; an analysis should read the file metadata rather than reproduce the table by hand.

The EMT is a derived vector-channel observable. Let $D_\mu = \frac{1}{2}(D_{+\mu} - D_{-\mu})$, let $P_{q,\tau}$ contain the spatial Fourier phase and the projector onto insertion time τ, and set $S_f = KD^{-1}K^\dagger$. The connected and disconnected contractions use the same local operator

$$B_{A\mu}(x) = \frac{1}{2}\bar{\chi}(x)\Gamma_A\left(\overrightarrow{D}_\mu - \overleftarrow{D}_\mu\right)\chi(x). \tag{20}$$

This convention is the flowed fermion EMT derivative used by Makino–Suzuki; the same two-sided twist-two derivative convention underlies lattice disconnected nucleon measurements such as arXiv:1310.6339 and nucleon spin/momentum decompositions such as DESY 17-086. After the closed-fermion-loop Wick sign, its exact disconnected trace is

$$L_{A\mu}(q,\tau) = -\frac{1}{2}\operatorname{Tr}[P_{q,\tau}\Gamma_A D_\mu S_f + D_\mu P_{q,\tau}\Gamma_A S_f]\,. \tag{21}$$

The old one-sided replacement $-\operatorname{Tr}[P_{q,\tau}\Gamma_A D_\mu S_f]$ is equal to Eq. (21) only when $P_{q,\tau}$ is constant along the derivative direction. It therefore fails for a spatial derivative when $q_\mu \neq 0$, and it fails for the temporal derivative at fixed τ even when $\mathbf{q} = 0$. Summing over all insertion times removes the latter obstruction, but a time-resolved three-point function cannot use that shortcut.

Production completes the second term without additional covariant derivatives. Using $D_\mu^\dagger = -D_\mu$, $S_f^\dagger = \gamma_5 S_f \gamma_5$, and

$$\Gamma_A^\sharp = \gamma_5 \Gamma_A^\dagger \gamma_5, \tag{22}$$

define $A_{A\mu}(q,\tau) = \operatorname{Tr}[P_{q,\tau}\Gamma_A D_\mu S_f]$. Then

$$L_{A\mu}(q,\tau) = -\frac{1}{2}\left[A_{A\mu}(q,\tau) - A_{A^\sharp\mu}(-q,\tau)^*\right]. \tag{23}$$

The raw PyQUDA basis is closed under this map; the partner and sign tables are stored as `gamma5_hermiticity_partner` and `gamma5_hermiticity_sign`. Internally the momentum projector contains the stable union of requested q and required $-q$, while the HDF5 `qext` axis remains exactly the requested list. Since QUDA returns $D_{+\mu} - D_{-\mu} = 2D_\mu$, the code is equivalent to

```
tmp = U_f.pure_gauge.covDev(eta, mu) - U_f.pure_gauge.covDev(eta, mu + 4)
field_A = contract("...ia,Aij,...ja->A...", xi.data.conj(), gamma_A, tmp.data)
right = project_q_and_minus_q(field_A)
L_D[:, mu] = -0.25 * (right[A,q] - conj(right[A_sharp,-q]))
```

3.4.1 Direct left derivative versus the production estimator

The gamma5-hermiticity completion and an explicit derivative on the noise are two different stochastic estimators of the same trace. Define

$$\xi_f = K\xi, \qquad \eta_f = KD^{-1}\xi. \tag{24}$$

An explicit right/left estimator of Eq. (21) is

$$\widehat{L}^{\mathrm{direct}}_{A\mu}(q,\tau) = -\frac{1}{2}\left[\xi_f^\dagger P_{q,\tau}\Gamma_A D_\mu \eta_f - (D_\mu \xi_f)^\dagger P_{q,\tau}\Gamma_A \eta_f\right]. \tag{25}$$

Because $D_\mu^\dagger = -D_\mu$ and $\mathbb{E}_\xi[\xi\xi^\dagger] = 1$,

$$\mathbb{E}_\xi\left[\widehat{L}^{\mathrm{direct}}_{A\mu}\right] = -\frac{1}{2}\,\mathrm{Tr}[P_{q,\tau}\Gamma_A D_\mu S_f + D_\mu P_{q,\tau}\Gamma_A S_f]\,. \tag{26}$$

This is useful as an independent reference but requires the extra applications $D_\mu\xi_f$.

Production instead estimates Eq. (23). With

$$\widehat{A}_{A\mu}(q,\tau) = \xi_f^\dagger P_{q,\tau}\Gamma_A D_\mu \eta_f, \tag{27}$$

it uses

$$\widehat{L}^{\gamma_5}_{A\mu}(q,\tau) = -\frac{1}{2}\left[\widehat{A}_{A\mu}(q,\tau) - \widehat{A}_{A^\sharp\mu}(-q,\tau)^*\right]. \tag{28}$$

The equality needed here is the trace identity

$$\mathrm{Tr}[D_\mu P_{q,\tau}\Gamma_A S_f] = -\,\mathrm{Tr}\Big[P_{-q,\tau}\Gamma_A^\sharp D_\mu S_f\Big]^*. \tag{29}$$

It uses hermiticity and cyclic permutation inside the trace. It is not a matrix identity inside a fixed-noise quadratic form: in general

$$(D_\mu\xi_f)^\dagger P_{q,\tau}\Gamma_A\eta_f \neq \left(\xi_f^\dagger P_{-q,\tau}\Gamma_A^\sharp D_\mu \eta_f\right)^* \tag{30}$$

for one random vector. Consequently the two estimators need not agree source by source, even though

$$\mathbb{E}_\xi[\widehat{L}^{\mathrm{direct}}] = \mathbb{E}_\xi[\widehat{L}^{\gamma_5}] = L. \tag{31}$$

Their difference is zero-mean stochastic noise. A finite-noise validation must therefore compare the paired ensemble difference with its statistical uncertainty; elementwise per-source equality is not a valid criterion.

The following is a historical check from the original implementation, not a validation of the new HP/multiplier sources or SC12. This was checked on S8T8 with 256 counter-based Z_4 sources, all 16 Gamma channels, all four derivative directions, every time slice, and $q = 0, \pm\hat{x}, \pm\hat{y}, \pm\hat{z}$. For Wilson–clover solutions at tolerance 10^{-15}, the global paired difference was 0.981 standard errors from zero; an identity-propagator control gave 0.978 standard errors. One-rank and four-rank means agreed at relative L_2 approximately 2.0×10^{-16}. The independent left/right calculation therefore found no systematic discrepancy, including at nonzero momentum, while confirming that the estimators fluctuate differently at finite noise.

For the EMT, $B_{\nu\mu} = L^D_{\gamma_\nu,\mu}$; the first tensor index is the gamma direction and the second is the derivative direction. The primitive is symmetrized only when constructing $T_{\mu\nu}$.

The raw matrices deliberately retain the historical PyQUDA bit-mask basis used by the qTMD and two-point workflows. A stored matrix `physical_from_pyquda` defines

$$\Gamma_A^{\mathrm{phys}} = \sum_B C_{AB}\Gamma_B^{\mathrm{raw}}. \tag{32}$$

With $\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4$, the raw and physical axial labels agree for `X5` and `Z5`, while

$$\Gamma^{\text{phys}}_{Y5} = -\Gamma^{\text{raw}}_{Y5}, \qquad \Gamma^{\text{phys}}_{T5} = -\Gamma^{\text{raw}}_{T5}. \tag{33}$$

The six raw tensor matrices satisfy

$$\Gamma^{\text{raw}}_{S\mu\nu} = \gamma_\mu\gamma_\nu = \frac{1}{2}[\gamma_\mu, \gamma_\nu] \quad (\mu < \nu). \tag{34}$$

If the Hermitian Euclidean convention $\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu]$ is desired, the saved tensor channel must be multiplied by i. Recording this distinction avoids silently assigning inconsistent axial signs or tensor phases in later form factor analyses.

3.5 Symmetrization

The EMT tensor is symmetrized after all unsymmetrized components have been accumulated:

$$L^q_{\mu\nu} \leftarrow \frac{1}{2}\left(L^q_{\mu\nu} + L^q_{\nu\mu}\right). \tag{35}$$

The code then sets both entries equal. The HDF5 writer stores only the upper triangle $\mu \le \nu$, with dataset names `T11`, `T12`, …, `T44`. The lower triangle should be reconstructed by symmetry if needed.

3.6 Noise Average and Volume Normalization

For N_{eff} effective sources, the averaged quark one-point loop is

$$L^q_{\mu\nu,\text{avg}}(q,t;t_f) = \frac{1}{V_3}\frac{1}{N_{\text{eff}}}\sum_{r=1}^{N_{\text{eff}}} L^{q,(r)}_{\mu\nu}(q,t;t_f). \tag{36}$$

This per-source averaging formula applies to undiluted sources. For SC12, channels are summed and probes averaged as described below. Production stores part means; raw primitive data are optional. The complete-base finalizer stores the primitive and symmetric EMT for each independent base, together with volume-normalized averages and SEM. In optional raw canonical output:

$$\texttt{raw/derivative_bilinear_pervec} \leftrightarrow L^{D,(r)}_{A\mu}(q,t;t_f), \tag{37}$$

$$\texttt{avg/Tmunu/Tij} \leftrightarrow L^q_{ij,\text{avg}}(q,t;t_f). \tag{38}$$

3.7 Identity Bilinear and Flowed-Noise Norm

The identity scalar bilinear and the flowed-source norm are

$$\chi^{(r)}_0(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\,\xi^\dagger_r(x;t_f)\,\eta_r(x;t_f), \tag{39}$$

$$\chi^{(r)}_1(q,t;t_f) = \sum_{\mathbf{x}} \Phi_q(x)\,\xi^\dagger_r(x;t_f)\,\xi_r(x;t_f). \tag{40}$$

χ_0 is already the identity channel of the complete local basis and is not duplicated. χ_1 checks the noise normalization. In code:

```
scalar_trace = local_bilinear[gamma_list.index("I")]

dot_xi_xi = contract("etzyxbc,etzyxbc->etzyx", xi.data.conj(), xi.data)
flowed_noise_norm = impose_P_Breit_slice(dot_xi_xi, phases_3pt)
```

Operator schema version 5 stores the latter separately. Production uses `shard_mean/flowed_noise_norm`; complete-base output uses `base/flowed_noise_norm` and `avg/flowed_noise_norm`. Optional raw output also has `raw/flowed_noise_norm_pervec`. Finalized base values and averages use the same V_3 normalization as the primitive bilinears.

3.8 Ringed-Fermion Kinetic Normalization

The flowed-noise norm is not the ringed-fermion normalization. The kinetic observable is already contained in the zero-momentum diagonal quark EMT components. For each effective stochastic source r, the implementation forms

$$K_r(t_f, \tau) = -\frac{2}{V_3} \sum_{\mu=1}^{4} L^q_{\mu\mu,r}(q = 0, \tau; t_f). \tag{41}$$

The two-sided definition is essential for the time-resolved $\mu = 4$ term. At $q = 0$, vector matrices obey $\Gamma_\mu^\sharp = -\Gamma_\mu$, so the standalone implementation obtains the same per-source quantity by combining the one-sided contraction with its complex conjugate. Only after summing all space-time points is the historical one-sided kinetic estimator equivalent in expectation. No new inversion, fermion-flow update, covariant derivative, or MPI gather is needed: Eq. (41) is evaluated directly from the four diagonal vector channels of `raw/derivative_bilinear_pervec`. The momentum list must contain exactly one $q = 0$ entry for every EMT quark measurement.

The complete-base file stores `base/ringed/kinetic`, `base/ringed/kinetic_spacetime` and corresponding `avg/` and `sem_real/`, `sem_imag/` views. Optional raw canonical output stores `derived/ringed/kinetic_pervec` and `derived/ringed/kinetic_spacetime`. Each file is published with one atomic rename. No nonlinear ringed factor is stored: a factor computed from one configuration is not the final ensemble normalization.

For configuration c, the stored spacetime average is

$$K_c(t_f) = \frac{1}{N_{\text{base}} N_p N_t} \sum_{b,k,\alpha,\tau} K_{b,k,\alpha,c}(t_f, \tau), \tag{42}$$

where N_p is the number of probes per base and α labels one undiluted channel or all 12 point spin–color channels. There is no division by the number of dilution channels. The analysis must first form the gauge-ensemble mean

$$K_{\text{code}}(t_f) = \langle K_c(t_f) \rangle_{\text{cfg}}\,, \tag{43}$$

and only then, for a single-flavor trace in fundamental $SU(3)$, evaluate

$$Z^{(f)}_{\text{ring}}(t_f) = \frac{-2N_c}{(4\pi)^2 t_f^2 K^{(f)}_{\text{code}}(t_f)}. \tag{44}$$

In particular, $\langle 1/K_c \rangle_{\text{cfg}} \neq 1/\langle K_c \rangle_{\text{cfg}}$ in general, so configuration-local inverse factors must not be averaged. If instead the kinetic expectation value has already been summed or averaged over N_f degenerate flavors, the numerator and denominator must use the same flavor convention. A quark EMT bilinear made from ringed fields receives one factor Z_{ring}, because the square roots from χ and $\bar{\chi}$ multiply to this single bilinear factor. The $t_f = 0$ output step is kept as an unflowed diagnostic and must be excluded from Eq. (44). The standalone `application/flowed_quark_ringed_norm` workflow remains available for dedicated kinetic-only high-statistics calculations. Its `RingedQuark1pt` class inherits the EMT counter-noise, base/HP scheduling, inversion, fermion-flow, sample-log, and shard runner, but supplies an independent four-vector-diagonal contraction. It therefore does not allocate the 16 local or 64 derivative EMT primitive arrays. Production resume trusts the fingerprinted base log without probing HDF5, so a logged base may be transferred immediately.

3.9 Gradient-Flow Schedule for Quarks

The quark workflow measures before flowing. Therefore step 0 is the unflowed observable. If $\epsilon =$ `flow_epsilon`, then

$$t_f(n) = n\,\epsilon, \qquad n = 0, \ldots, \texttt{flow_steps}. \tag{45}$$

The first flow interval is internally split into ten smaller updates:

```
for step in range(Nsteps + 1):
    measure primitive bilinears and flowed_noise_norm
    if step == 0:
        flow [xi, eta] for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow [xi, eta] for 1 step of stepsize
```

The stochastic source ξ and solution η are flowed together as one `MultiField`. This keeps both fields on the same flowed gauge background. After all flow steps are measured, finalization derives the ringed kinetic algebraically from the $q = 0$ vector derivative diagonals: per independent base in `.bases.h5`, or per source in optional raw canonical output.

3.10 Four-Dimensional Fermion Flow and the Time-Slice Projector

Let $K(t_f)$ denote the gauge-covariant fermion-flow kernel from flow time zero to t_f. Suppressing source labels, the fields measured by the code are

$$\xi_f = K(t_f)\xi, \qquad \eta_f = K(t_f)D^{-1}\xi. \tag{46}$$

The fermion flow is four dimensional: its covariant Laplacian couples all four Euclidean directions, including time. Its characteristic smoothing radius is approximately $\sqrt{8t_f}$. This radius is a diffusion scale, not a strict compact-support boundary.

Let P_τ project the measured output field onto the absolute Euclidean time slice τ, while leaving spatial, spin, and color indices unchanged. For an EMT kernel Γ on the flowed gauge background, one stochastic measurement can be written as

$$\begin{aligned}\widehat{L}(\tau, t_f) &= \xi_f^\dagger P_\tau \Gamma \eta_f \\ &= \xi^\dagger K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}\xi.\end{aligned} \tag{47}$$

Using the full-volume noise identity in Eq. (4),

$$\begin{aligned}\mathbb{E}_\xi[\widehat{L}(\tau, t_f)] &= \mathrm{Tr}\Big[K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}\Big] \\ &= \mathrm{Tr}\Big[P_\tau \Gamma K(t_f) D^{-1} K^\dagger(t_f)\Big].\end{aligned} \tag{48}$$

The projector P_τ means that the physical observable remains a spatial sum at a resolved insertion time; Eq. (48) is not a sum over insertion time. Nevertheless, the initial source must cover the full four-dimensional vector space because $K^\dagger P_\tau \Gamma K D^{-1}$ connects the measured time slice to source components at other times.

Equivalently, define

$$A_\tau = K^\dagger(t_f) P_\tau \Gamma K(t_f) D^{-1}. \tag{49}$$

For finite noise count, the off-diagonal stochastic contribution is

$$\widehat{L}(\tau, t_f) - L(\tau, t_f) = \sum_{i \neq j} \xi_i^* (A_\tau)_{ij} \xi_j. \tag{50}$$

The left index is constrained by the flowed time-slice operator, but the right index can originate from any Euclidean time. These terms vanish in the noise expectation; their finite-sample size is a variance question.

3.11 Why a Single Initial-Time Projector Is Not the Current Estimator

Suppose one replaces the full-volume source by only one time-diluted source $\xi_d = P_d\xi$. At nonzero flow time its expectation value is

$$\mathbb{E}_\xi\left[\xi^\dagger P_d K^\dagger P_\tau \Gamma K D^{-1} P_d \xi\right] = \mathrm{Tr}\left[P_d K^\dagger P_\tau \Gamma K D^{-1} P_d\right], \tag{51}$$

which is generally not Eq. (48). Four-dimensional fermion flow spreads the endpoint time slice over several initial-time components, so choosing only $d = \tau$ omits contributions.

Time dilution is not invalid in principle. For a complete orthogonal family

$$\sum_d P_d = \mathbf{1}, \qquad P_d P_{d'} = \delta_{dd'} P_d, \tag{52}$$

summing all diluted estimators reconstructs the full trace:

$$\sum_d \mathrm{Tr}[P_d A_\tau P_d] = \mathrm{Tr}\, A_\tau. \tag{53}$$

It requires all projectors and additional inversions. The present EMT workflow therefore uses one full-volume base source per stochastic solve and retains all absolute insertion times. In the unflowed limit $t_f = 0$, where $K(0) = \mathbf{1}$, a local loop at τ can be obtained from the matching exact-time projector; this special case does not justify dropping the other projectors for the finite-flow measurements stored in the same file.

4 Probing, Dilution, and Source Bookkeeping

EMT supports `zn`, `hierarchical_probing`, and `multiplier_probing`. Each method starts from a full-volume counter-based random base ξ_b. Probe and dilution channels within a base are correlated; only independently randomized complete bases are statistical samples.

4.1 Complete Nested Hierarchical Probing

HP multiplies the same base by deterministic site-only signs,

$$\xi_{b,k}(x) = h_k(x)\xi_b(x), \qquad h_k(x) \in \{-1, +1\}. \tag{54}$$

The only supported ordering is

`nested_xyzt_checkerboard_v1`

It is defaulted in shared Python code; shell wrappers only forward explicit ordering overrides. Old Gray/interleaved generators and arbitrary power-of-two HP counts are not supported. Complete levels are:

HP count	q in $N_p = 2q^4$	Cancellation distance	Extent divisor
2	1	1	2
32	2	3	4
512	4	7	8

Every global extent must be divisible by the indicated divisor; it need not be a power of two. A partial HP prefix is a checkpoint, not another independent base. The source remains full-volume; HP is not time dilution.

4.2 Multiplier Fourier Probing

EMT accepts a shape-specific `multiplier_plan` dictionary or JSON path; the application forwards `EMT_1PT_MULTIPLIER_PLAN`. The plan records version, four global extents, distance, number of colors M, and four integer coefficients σ. Rank zero reads and validates it, then broadcasts the canonical plan or an error. The actual lattice shape must match the plan. For a site color

$$c(x) = \sigma \cdot x \pmod{M}, \tag{55}$$

the complete Fourier basis is

$$\xi_{b,k}(x) = e^{2\pi i k c(x)/M} \xi_b(x), \qquad 0 \le k < M. \tag{56}$$

All M probes reuse the same random base and their estimates are averaged with weight $1/M$. These are unit-modulus Fourier sources, not unscaled color masks; a mask estimator would instead require summation. There is no runtime plan search or automatic change of the color count. Supplied plans and their search evidence are in `application/EMT_disconnected_1pt/probing_plans`. Color-count search results do not establish minimum wall time or estimator variance.

4.3 Point Spin–Color Dilution and Cost

EMT optionally accepts `spin_color_dilution=point` through its Python measurement API. The production CLI does not expose an SC12 switch. For orthogonal spin–color projectors P_α, $\alpha = 0, \ldots, 11$,

$$\xi_{b,k,\alpha} = P_\alpha \xi_{b,k}, \qquad L_b = \frac{1}{N_p} \sum_{k=0}^{N_p-1} \sum_{\alpha} L_{b,k,\alpha}. \tag{57}$$

The channels are summed, not averaged. The default `none` uses one undiluted channel. Thus

$$N_{\text{eff}} = N_{\text{base}} N_p d, \qquad d = \begin{cases} 1 & \text{undiluted,} \\ 12 & \text{point dilution.} \end{cases} \tag{58}$$

Here $N_p = 1$ for Z_n, the complete HP count for HP, or M for multiplier. `EMT_1PT_N_VEC` counts bases, not solves. Standalone ringed and qTMD retain only Z_n/complete HP without SC12 or multiplier; the EMT options must not be assumed available there.

4.4 Stored Metadata and Raw Bookkeeping

The single stored probe count is `vectors_per_base`. Metadata also records `noise_scheme` and explicit `spin_color_dilution`; HP stores `hp_ordering`, while multiplier stores `multiplier_plan_json`. `hp_num_vectors` remains an HP-only input, not a second stored count. The total independent count is `n_base_noise`; the solve count is `effective_n_inversions`. Optional raw bookkeeping uses

```
raw/base_noise_index
raw/probe_index
raw/spin_index       # point dilution only
raw/color_index      # point dilution only
```

The effective source index is reconstructed as

$$r = (bN_p + k)d + \alpha, \tag{59}$$

with $\alpha = 3s + c$ for SC12 and zero for undiluted sources. It is not stored as another dataset. For one undiluted HP2 base the base indices are $[0, 0]$, probe indices $[0, 1]$, and source indices $[0, 1]$.

5 Quark HDF5 Layout

Production writes complex128 part means, with raw per-source data optional. Default finalization requires those means and writes `<canonical-tag>.bases.h5`. Its volume-normalized datasets include

```
base/base_noise_index
base/local_bilinear           [Nb,16,Nq,Nflow,Nt]
base/derivative_bilinear      [Nb,16,4,Nq,Nflow,Nt]
base/flowed_noise_norm        [Nb,Nq,Nflow,Nt]
base/Tmunu/T11 ... T44        [Nb,Nq,Nflow,Nt]
base/ringed/kinetic           [Nb,Nflow,Nt]
base/ringed/kinetic_spacetime [Nb,Nflow]
avg/...
sem_real/...
sem_imag/...
```

Here $N_b = N_{\text{base}}$ and $N_{\text{flow}} = N_{\text{steps}} + 1$. SEM is computed across independent complete bases, separately for real and imaginary parts; it is NaN for one base. These are fixed-configuration stochastic errors, not gauge-ensemble uncertainties.

For optional raw diagnostics, measure with `--save-raw-per-vector`. Then use the Python finalizer's `--raw-output` option to also write `<canonical-tag>.h5`. The latter is an optional diagnostic file, not the default statistical input. Its raw datasets are

$$\texttt{raw/local_bilinear_pervec} : [N_{\text{eff}}, 16, N_q, N_{\text{steps}} + 1, N_t], \tag{60}$$

$$\texttt{raw/derivative_bilinear_pervec} : [N_{\text{eff}}, 16, 4, N_q, N_{\text{steps}} + 1, N_t], \tag{61}$$

$$\texttt{raw/flowed_noise_norm_pervec} : [N_{\text{eff}}, N_q, N_{\text{steps}} + 1, N_t]. \tag{62}$$

The averaged datasets are

$$\texttt{avg/local_bilinear} : [16, N_q, N_{\text{steps}} + 1, N_t], \tag{63}$$

$$\texttt{avg/derivative_bilinear} : [16, 4, N_q, N_{\text{steps}} + 1, N_t], \tag{64}$$

$$\texttt{avg/Tmunu/T11}, \texttt{T12}, \ldots, \texttt{T44} : [N_q, N_{\text{steps}} + 1, N_t], \tag{65}$$

$$\texttt{avg/flowed_noise_norm} : [N_q, N_{\text{steps}} + 1, N_t]. \tag{66}$$

Only $\mu \leq \nu$ tensor components are written in the averaged `avg/Tmunu` group.

Both canonical outputs are source independent. The optional raw name is:

`EMTquarkLoop/<lat>.EMTquarkLoop.<cfg>.<ama>.<sm>.h5`.

It stores every absolute insertion time and is reused for all hadron two-point source times on the same configuration. Before forming a disconnected three-point product, analysis applies `roll(time_axis,-source_t)` to both quark and gluon loops. Thus output index τ_{rel} represents $\tau_{\text{abs}} = (t_0 + \tau_{\text{rel}}) \bmod N_t$.

The embedded kinetic-only schema is

```
derived/ringed/kinetic_pervec
derived/ringed/kinetic_spacetime
```

These `derived/ringed` paths belong to optional raw output; the default base file uses `base/ringed`, `avg/ringed` and SEM groups. EMT defaults to undiluted spin–color noise and optionally supports SC12. The group attributes record the unique zero-momentum index, the $-2/V_3$ relation, and that ringed factors are not stored. `ringed_factors_stored=False` states that nonlinear normalization is an ensemble-analysis responsibility. The counter-noise configuration, stream, algorithm, HP ordering, Dirac parameters, gauge preprocessing, boundary condition, flow schedule, and exact EMT-to-kinetic relation are recorded alongside it.

5.1 Raw-Bilinear Analysis Example

The vector channels occur at Gamma positions

$$(X, Y, Z, T) = (3, 5, 7, 1). \tag{67}$$

For the already averaged and volume-normalized primitive, every EMT component can be reconstructed without reading the large source axis:

```
import h5py
import numpy as np

with h5py.File(path, "r") as h5:
    labels = [x.decode() for x in h5["gamma_list"][...]]
    vector = [labels.index(x) for x in ("X", "Y", "Z", "T")]
    D = h5["avg/derivative_bilinear"][...]
    # B axes are [nu,mu,q,flow,t_abs].
    B = np.take(D, vector, axis=0)
    T = 0.5 * (B + np.swapaxes(B, 0, 1))
    np.testing.assert_allclose(T[3,3], h5["avg/Tmunu/T44"][...])
```

Here `T[mu,nu]` has axes `[q,flow,t_abs]` after the two tensor indices. The division by `volume_norm` must not be repeated because it is already included in `avg/derivative_bilinear`. For stochastic-error analysis, read the normalized `base/Tmunu` and `base/ringed` datasets from `.bases.h5` directly. The ringed comparison, convergence and proton T44 helpers do not regroup raw vectors or read old HP16/HP256 files. Their solve costs come from the stored probe count and dilution factor. Optional raw files remain useful for per-source diagnostics, not independent-probe SEM.

The same raw basis also supports other observables. First apply `physical_from_pyquda` on the Gamma axis. Symmetrizing the physical `X5,Y5,Z5,T5` derivative channels yields

$$A_{\mu\nu} = \frac{1}{2}\left(L^D_{\gamma_\mu\gamma_5,\nu} + L^D_{\gamma_\nu\gamma_5,\mu}\right). \tag{68}$$

Selecting the six local tensor channels gives the raw tensor currents; multiply by i only for the Hermitian $i[\gamma_\mu, \gamma_\nu]/2$ convention. A complete portable example, including connected pion and proton axis orders, is in `docs/EMT_gamma_and_raw_bilinears.md`.

5.2 Historical Raw-File Storage Example

The following measured sizes describe the original raw-output implementation, not a new mean-only or complete-base run. The values are retained as a historical storage example. Default production storage now scales with parts, and finalized base observables with independent bases. For this raw example, all large arrays are uncompressed. Ignoring the small averaged and metadata datasets, the source-scaling payload has the ratio

$$\texttt{derivative:local:flowed_noise_norm} = 64 : 16 : 1. \tag{69}$$

The embedded ringed kinetic adds one $q = 0$ channel, or $1/N_q$ unit on this scale. In the measured S8T8 file with $N_{\text{eff}} = 8192$, $N_q = 9$, two flow times, and $N_t = 8$, the complete file is 1,531,264,312 bytes:

Part	Bytes	Complete-file share
`raw/derivative_bilinear_pervec`	1,207,959,552	78.886%
`raw/local_bilinear_pervec`	301,989,888	19.722%
`raw/flowed_noise_norm_pervec`	18,874,368	1.233%
`derived/ringed/kinetic_pervec`	2,097,152	0.137%
All averaged primitives and ten `Tmunu` datasets	209,664	0.014%
Bookkeeping, basis metadata, and HDF5 overhead	remainder	0.008%

Thus the 64 per-vector derivative channels determine the file size; the ten convenience `avg/Tmunu` datasets are negligible.

5.3 Base-Oriented Production, Checkpoints, and Finalization

Production has one shard path, with mandatory means and optional raw data:

```
shard_mean/local_bilinear
shard_mean/derivative_bilinear
shard_mean/flowed_noise_norm
```

Parts contain at most `block_interval_solves` inversions (default 64), without splitting an SC12 group. Undiluted HP512 therefore produces eight 64-probe parts; HP2, HP32 and plain noise each produce one part per base. A partial part is a storage/checkpoint unit, not an independent estimator. For example,

```
<stem>.base000003.part0001.probe0064-0127.h5
```

Rank zero writes each part to a unique temporary path and atomically renames it. Only after all parts of a base close does rank zero append an exact `baseXXXXXX` line to the fingerprinted sample log. Resume trusts this log without probing HDF5; transferred logged parts need not remain locally. An unlogged base is recomputed in full. Jobs may own nonoverlapping base ranges; overlapping jobs are unsupported.

New format identities are explicit:

```
shard_schema = disconnected_1pt_base_parts_v4
sample-log header = disconnected_sample_log_v2
source_bookkeeping_schema = base_probe_v2
# For SC12 instead:
source_bookkeeping_schema = base_probe_spin_color_v3
```

Use a fresh output/setup identity for every new-format run, including Z_n. Old files/logs are rejected, not migrated. Changing the probe count, plan, ordering, dilution or raw-storage mode also requires a new identity. Planned base counts may be extended; do not put that count in the setup tag.

The default serial finalizer requires all requested bases and mandatory means. It validates complete coverage, metadata, finite means and optional raw/mean agreement. It weights parts by their actual probe counts, sums SC12 channels and averages only probes, then derives normalized base EMT and kinetic values. There is no raw-only fallback. Streaming mean/variance accumulation gives real/imaginary SEM; an atomic rename publishes the complete base file.

From the application directory, after matching the data directory, lattice tag, setup tag, configuration and total base count to the measurement:

```
bash run_finalize_quark_1pt.sh --config_num 1000
# Optional raw diagnostics, requiring raw measurement:
python Pyquda_EMT_disconnected_finalize_quark_1pt.py \
  --config_num 1000 --raw-output
```

The shell wrapper performs default base finalization only. The Python `--raw-output` option writes both base statistics and raw diagnostics; there is no `--base-statistics` switch. Standalone ringed and qTMD keep their own raw finalizers and must not be described as mean-only EMT workflows.

One runtime detail is essential: evaluating a flowed contraction loads the flowed gauge into QUDA's global resident gauge state. Before inverting the next stochastic source, the code therefore reloads the original unflowed input gauge. The input Python gauge object is unchanged, but omitting this resident-state restore makes later inversions use the wrong gauge and can trigger precision mismatch failures.

The initial full `dirac.loadGauge(U)` is itself the first valid resident original-gauge state. Therefore the first inversion batch does not perform a redundant thin reload. After the first flowed-gauge context, every later batch uses `thin_update_only=True` exactly once before inversion.

Canonical quark and gluon loops are projected with Fourier origin $o = (0, 0, 0, 0)$ and retain absolute lattice time. Their HDF5 provenance records the global lattice size, `momentum_phase_origin`, phase convention, and time convention. Combining a loop with a hadron source x_0 uses

$$L_{x_0}(\boldsymbol{q}, \tau) = \exp\left[-2\pi i \sum_{j=1}^{3} q_j \frac{x_{0j} - o_j}{L_j}\right] L_o(\boldsymbol{q}, (t_0 + \tau) \bmod N_t)\,. \tag{70}$$

Thus source-relative analysis requires both spatial rephasing and periodic time rolling. Old loops without this provenance are rejected rather than silently interpreted with an assumed origin.

For the optional raw canonical path, downstream diagnostic names include:

```
gamma_list
gamma_pyquda_ids
gamma_matrices
physical_from_pyquda
gamma5_hermiticity_partner
gamma5_hermiticity_sign
derivative_directions
raw/local_bilinear_pervec
raw/derivative_bilinear_pervec
raw/flowed_noise_norm_pervec
raw/base_noise_index
raw/probe_index
avg/Tmunu/T11
avg/Tmunu/T12
avg/Tmunu/T22
avg/Tmunu/T33
avg/Tmunu/T44
avg/local_bilinear
avg/derivative_bilinear
avg/flowed_noise_norm
derived/ringed/kinetic_pervec
derived/ringed/kinetic_spacetime
noise_scheme
vectors_per_base
spin_color_dilution
effective_n_inversions
```

Important attributes include

Attribute	Meaning
`measurement`	`quark_1pt`.
`flow_type`, `flow_epsilon`, `flow_steps`, `flow_times`	Gradient-flow schedule.
`qext`	Momentum labels used by downstream analysis.
`loop_provenance_schema`, `momentum_phase_origin`	Versioned Fourier-origin convention; current production is v2.
`spatial_momentum_phase_convention`	$\exp[+2\pi i\, \boldsymbol{q} \cdot (\boldsymbol{x} - \boldsymbol{o})/\boldsymbol{L}]$.
`volume_norm`	V_3, the spatial-volume normalization.
`derived_emt_upper_triangle_only`	Whether only $\mu \leq \nu$ components are written.
`n_base_noise`, `effective_n_inversions`	Stochastic source counts.
`noise_scheme`, `vectors_per_base`	Source scheme and probe count.
`spin_color_dilution`	`none` or `point`.
`hp_ordering` / `multiplier_plan_json`	Present for HP / multiplier respectively.
`n_zn`, `config_num`, `noise_stream`	Counter-noise order, configuration key, and independent stream salt.
`noise_generator`, `noise_counter_order`	Fixed hash algorithm and the fields entering its counter.
`operator_normalization`, `renormalization_applied`	Whether the saved quark loop has already been renormalized.
`gamma5_hermiticity_reconstruction`, `derivative_closed_fermion_loop_sign_included`	The left derivative is reconstructed and the saved loop contains one Wick minus.

6 Gluon One-Point Workflow

The gluon workflow is independent of stochastic quark sources. It constructs a flowed gauge-field strength and then measures a spatially Fourier-projected gluon EMT building block.

6.1 Clover Field Strength

For each independent plane (μ, ν), the code constructs the four oriented clover plaquettes

$$\mu \to \nu \to -\mu \to -\nu, \tag{71}$$
$$\nu \to -\mu \to -\nu \to \mu, \tag{72}$$
$$-\mu \to -\nu \to \mu \to \nu, \tag{73}$$
$$-\nu \to \mu \to \nu \to -\mu. \tag{74}$$

Let $Q_{\mu\nu}(x)$ be their sum. The anti-Hermitian clover field is

$$A_{\mu\nu}(x) = \frac{1}{8}\left[Q_{\mu\nu}(x) - Q^{\dagger}_{\mu\nu}(x)\right]. \tag{75}$$

The code projects this matrix to the traceless $\mathfrak{su}(3)$ algebra:

$$A^{\mathrm{tl}}_{\mu\nu}(x) = A_{\mu\nu}(x) - \frac{1}{N_c}\,\mathrm{Tr}_c[A_{\mu\nu}(x)]\,\mathbf{1}. \tag{76}$$

Finally, the Euclidean field-strength convention used by the implementation is

$$F_{\mu\nu}(x) = -i\,A^{\mathrm{tl}}_{\mu\nu}(x). \tag{77}$$

Only the six independent planes are computed, and the antisymmetric entries are filled with

$$F_{\nu\mu}(x) = -F_{\mu\nu}(x). \tag{78}$$

The traceless projection is a field-strength projection. It is not a traceless projection of the final EMT tensor.

6.2 Implemented Gluon EMT Loop

The implemented gluon one-point building block is

$$L^g_{\mu\nu}(q,t;t_f) = \frac{2}{V_3}\sum_{\mathbf{x}} \Phi_q(x) \sum_{\substack{\rho=1 \\ \rho\neq\mu,\nu}}^{4} \mathrm{Tr}_c[F_{\mu\rho}(x;t_f)\, F_{\nu\rho}(x;t_f)]\,. \tag{79}$$

This maps directly to the code in `EMTDisconnectedGluon1pt.flowed_1pt`:

```
for mu in range(4):
    for nu in range(mu, 4):
        tmp = zeros(...)
        for rho in range(4):
            if rho == mu or rho == nu:
                continue
            tmp += contract("...ab,...ba->...", F[mu][rho], F[nu][rho])
        Tmunu_t[mu, nu, :, step, :] += 2.0 * slice_t
Tmunu_t /= Ns3
```

The factor 2 is explicit in the code. The division by V_3 is applied after all spatial Fourier sums have been accumulated.

6.3 Gradient-Flow Schedule for Gluons

As in the quark workflow, step 0 is measured before flow and therefore is the unflowed gauge observable. After each measurement, the gauge field is advanced by Wilson or Symanzik flow:

```
for step in range(Nsteps + 1):
    measure F and Tmunu
    if step == 0:
        flow gauge for 10 substeps of stepsize/10
    elif step < Nsteps:
        flow gauge for 1 step of stepsize
```

The production implementation deliberately keeps this materialize-and-advance schedule. A persistent resident gauge-flow context was tested for Wilson and Symanzik flow with one and ten output steps. It reproduced every saved component exactly, but after separately prewarming both paths its end-to-end speed changed by only a few percent and was not consistently faster in both execution orders. It therefore did not meet the ten-percent integration threshold. This avoids adding resident-state lifetime logic without a stable performance benefit.

7 Gluon HDF5 Layout

The gluon writer is `save_emt_gluon_1pt_hdf5`. Unlike the quark file, there are no stochastic raw per-vector datasets. The output contains only volume-normalized upper-triangle tensor components:

$$\texttt{Tmunu/T11}, \texttt{T12}, \ldots, \texttt{T44} : [N_q, N_{\mathrm{steps}}+1, N_t]. \tag{80}$$

The canonical gluon loop is source independent,

`EMTgluonLoop/<lat>.EMTgluonLoop.<cfg>.<ama>.<sm>.h5`,

and is reused for all hadron source positions on the configuration. The shared quark and gluon wrappers use the same `EMT_1PT_FLOW_EPSILON`; its production default is (0.207936), so equal flow indices refer to equal flow times. Important attributes include

Attribute	Meaning
`measurement`	`gluon_1pt`.
`config_num`	Gauge configuration encoded by the canonical file.
`flow_type`, `flow_epsilon`, `flow_steps`, `flow_times`	Gradient-flow schedule.
`qext`, `pf`, `p_2pt`	Momentum labels used by downstream analysis.
`volume_norm`	V_3, the spatial-volume normalization.
`upper_triangle_only`	Whether only $\mu \leq \nu$ components are written.

8 Disconnected Combination with Proton Two-Point Functions

The quark and gluon one-point loops are hadron independent. They become a proton disconnected three-point building block only after being correlated with the proton two-point function on the gauge ensemble. For a proton two-point function $C_2^p(p_f, t)$, define

$$C_{3,\mu\nu}^{p,\mathrm{disc}}(p_f, p_i; t, \tau; t_f) = \langle C_2^p(p_f, t)\, L_{\mu\nu}(q, \tau; t_f)\rangle_{\mathrm{cfg}} - \langle C_2^p(p_f, t)\rangle_{\mathrm{cfg}}\, \langle L_{\mu\nu}(q, \tau; t_f)\rangle_{\mathrm{cfg}}\,, \tag{81}$$

with

$$q = p_f - p_i. \tag{82}$$

The subtraction term removes the vacuum expectation value of the one-point loop. This subtraction must be done at the ensemble-analysis level using the same configuration set as the proton two-point function.

A basic ratio is

$$R_{\mu\nu}^{p,\mathrm{disc}}(t, \tau; t_f) = \frac{C_{3,\mu\nu}^{p,\mathrm{disc}}(p_f, p_i; t, \tau; t_f)}{C_2^p(p_f, t)}. \tag{83}$$

Kinematic projectors, EMT mixing coefficients, and gradient-flow matching factors are applied later in the physics analysis.

9 End-to-End Proton Diagnostic Workflow

The application directory

`application/EMT_disconnected_1pt/perlmutter`

contains a quark-only diagnostic workflow:

1. measure undiluted quark shards with `--save-raw-per-vector`;
2. run the Python finalizer with `--raw-output`;
3. measure the proton two-point function C_2^p;
4. combine C_2^p with the raw canonical quark loop.

The builder reads neither `.bases.h5` nor SC12 raw files. A separate gluon measurement is optional and used only with `--include_gluon`. Ordinary complete-base convergence analysis does not need raw data or this builder.

For the quark loop, the builder reads `raw/derivative_bilinear_pervec`, reconstructs the symmetric vector EMT, and forms cumulative stochastic averages

$$\overline{L}^{q,N}_{\mu\nu}(q,t;t_f) = \frac{1}{V_3}\frac{1}{N}\sum_{r=1}^{N} L^{q,(r)}_{\mu\nu}(q,t;t_f), \qquad N = 1,\ldots,N_{\mathrm{eff}}. \tag{84}$$

This is the quantity used to check whether the stochastic estimator changes as more sources are included. Partial HP/multiplier prefixes are diagnostics, not independent complete-base samples and not valid additional SEM units.

For a single configuration, the builder can only form the unsubtracted proxy

$$C^{\mathrm{unsub},N}_{3,\mu\nu}(t_{\mathrm{sep}},q,t;t_f) = C^{p}_{2}(t_{\mathrm{sep}})\,\overline{L}^{q,N}_{\mu\nu}(q,t;t_f). \tag{85}$$

This object is useful for checking shapes, source-count dependence, time alignment, and HDF5 plumbing. It is not the physical disconnected correlator, because the vacuum subtraction in Eq. (81) requires an ensemble average.

With multiple configurations, the builder applies the ensemble subtraction for each cumulative source count:

$$C^{\mathrm{disc},N}_{3,\mu\nu} = \left\langle C^{p}_{2}\,\overline{L}^{q,N}_{\mu\nu}\right\rangle_{\mathrm{cfg}} - \langle C^{p}_{2}\rangle_{\mathrm{cfg}}\left\langle \overline{L}^{q,N}_{\mu\nu}\right\rangle_{\mathrm{cfg}}, \tag{86}$$

$$R^{\mathrm{disc},N}_{\mu\nu} = \frac{C^{\mathrm{disc},N}_{3,\mu\nu}}{\langle C^{p}_{2}\rangle_{\mathrm{cfg}}}. \tag{87}$$

The gluon loop has no stochastic source axis, so the builder forms the same unsubtracted and vacuum-subtracted objects without a cumulative N index.

The diagnostic merger output contains:

```
C2
quark/source_count
quark/loop_cumulative
quark/C3_unsubtracted_cumulative
quark/C3_disc_cumulative     # only if Ncfg >= 2
quark/R_disc_cumulative      # only if Ncfg >= 2
gluon/loop
gluon/C3_unsubtracted
gluon/C3_disc                # only if Ncfg >= 2
gluon/R_disc                 # only if Ncfg >= 2
summary/quark_source_count
summary/quark_T44_q0_flow0_loop_norm
summary/quark_T44_q0_flow0_unsub_ratio_proxy
```

10 How to Read Smoke Tests

A single `S8T32` test gauge checks implementation conventions, not physics signals. Useful smoke-test checks are:

- the code runs on the intended backend;
- `flow_times` contains step 0 and the requested flow times;

- `derived_emt_upper_triangle_only=True`;
- default output has normalized `base/`, `avg/`, `sem_real/`, and `sem_imag/` datasets;
- one complete base has NaN SEM; SC12 costs 12 solves per probe;
- when raw diagnostics are requested, datasets have the expected $[N_{\text{eff}}, 16, N_q, N_{\text{steps}}+1, N_t]$, $[N_{\text{eff}}, 16, 4, N_q, N_{\text{steps}}+1, N_t]$, and $[N_{\text{eff}}, N_q, N_{\text{steps}}+1, N_t]$ shapes;
- HP runs have consistent `raw/base_noise_index` and `raw/probe_index`, from which the effective source coordinate is reconstructed;
- gluon datasets contain `Tmunu/T11` through `T44` and no stochastic quark primitive datasets.

11 Validation Scope and Current Limitations

CPU/mock regressions exercise nested HP cancellation, validated multiplier plans, SC12 normalization, resume, complete-base aggregation and analysis. They do not validate QUDA kernels, GPU queues or MPI communication. New probing GPU/MPI validation remains pending; the historical operator checks earlier in this note must not be presented as new probing validation. The dedicated `tests/check_disconnected_probing_gpu.py` entry supports HIP/CuPy and SYCL/dpnp, requires a new output directory, and can check source hashes across decompositions and complete EMT outputs. It submits no job. See `PROBING.md` for commands and matching environment setup.

Multiplier probing is implemented, not a future placeholder. The supplied plans are shape-specific, and their coloring/search evidence does not establish minimum estimator variance or optimal production wall time. Compare methods at matched total inversion cost and with enough independent complete bases. No automatic old-file migration or probing-layer extension is provided.